AI Learning Companion

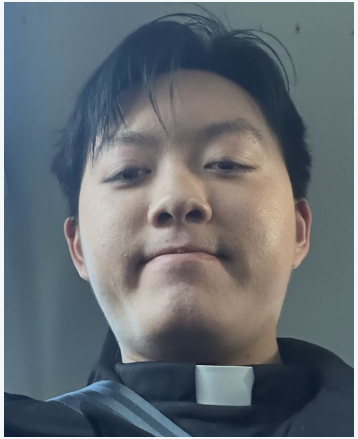
Team 7 Needfinding

Team Members

Ayala Wang

‘26

San Jose, CA



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Las Vegas, NV

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Project Name: Reason Lab

“Better Learning Starts With You “

Problem Overview

- Audience / Users
  - College + High School Students dealing w/ complex and difficult to understand coursework

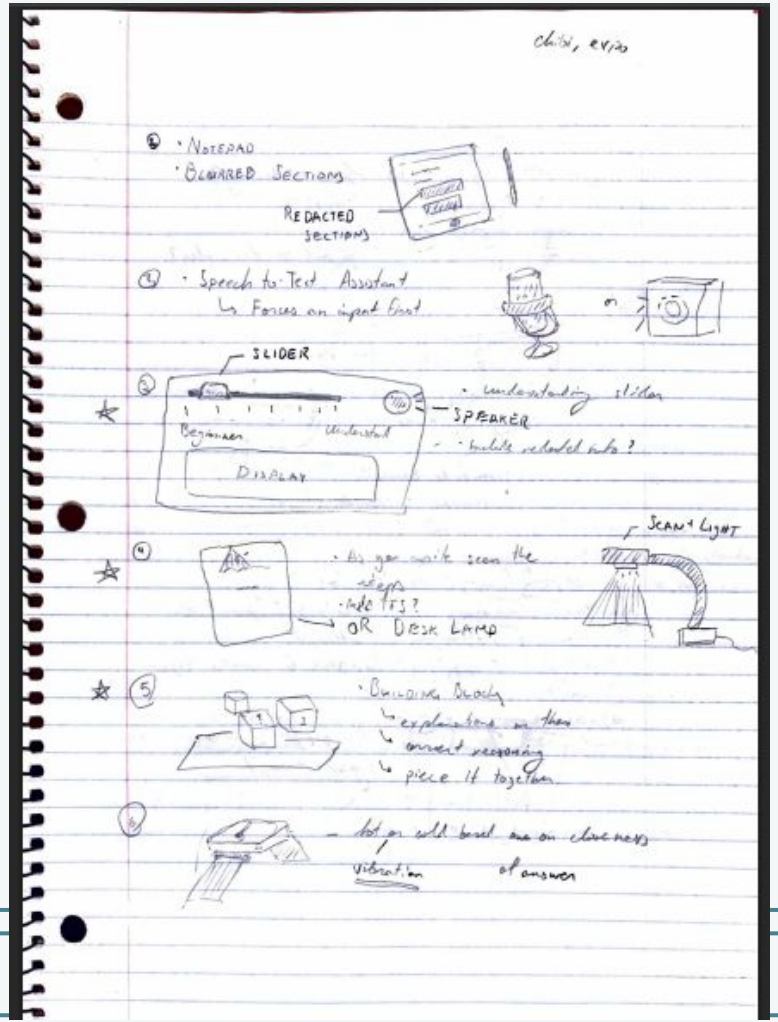
- Problem
  - Nowadays with AI so readily available, many are using it as a shortcut.
  - This use of AI is taking away from in-depth learning, and can

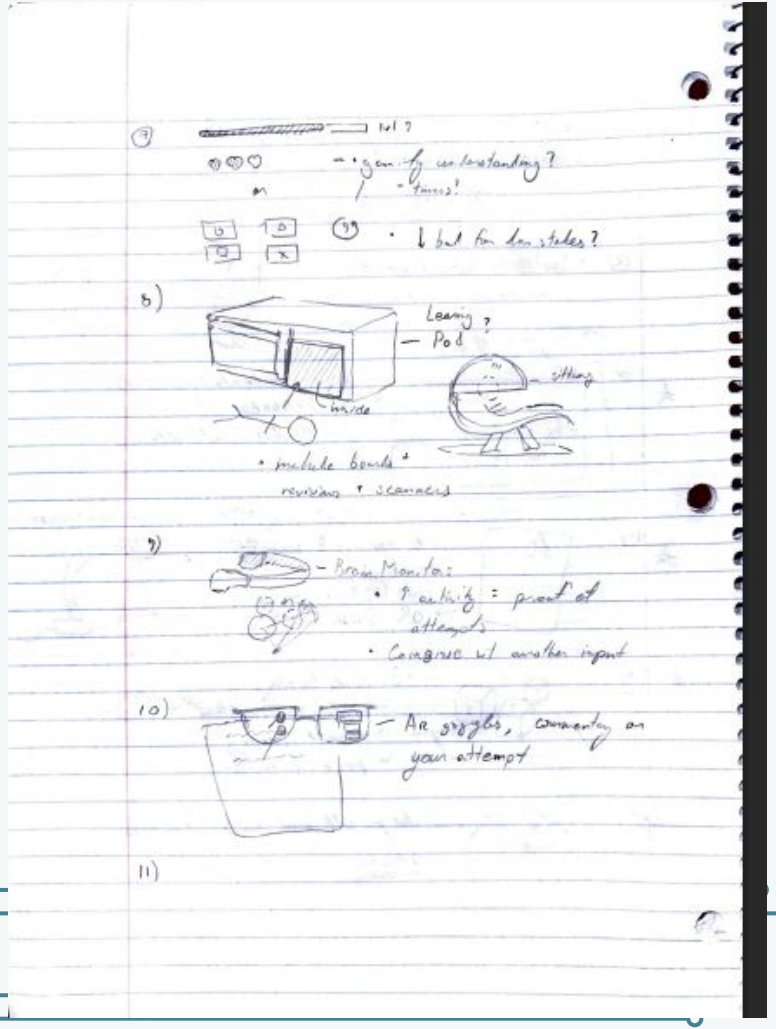
What is our solution?

- The value proposition of Reason Lab is to enforce active learning by requiring users to attempt a solution before receiving assistance.
- Solution:
  - Makes its users strengthen their own reasoning rather than an instant solution
  - Will give adaptive guidance and guide the user, forcing a more durable understanding of material
  - Through adaptive, visual guidance, ReasonLab responds to user attempts and misconceptions, helping learners refine their thinking and build a more durable conceptual understanding over time.

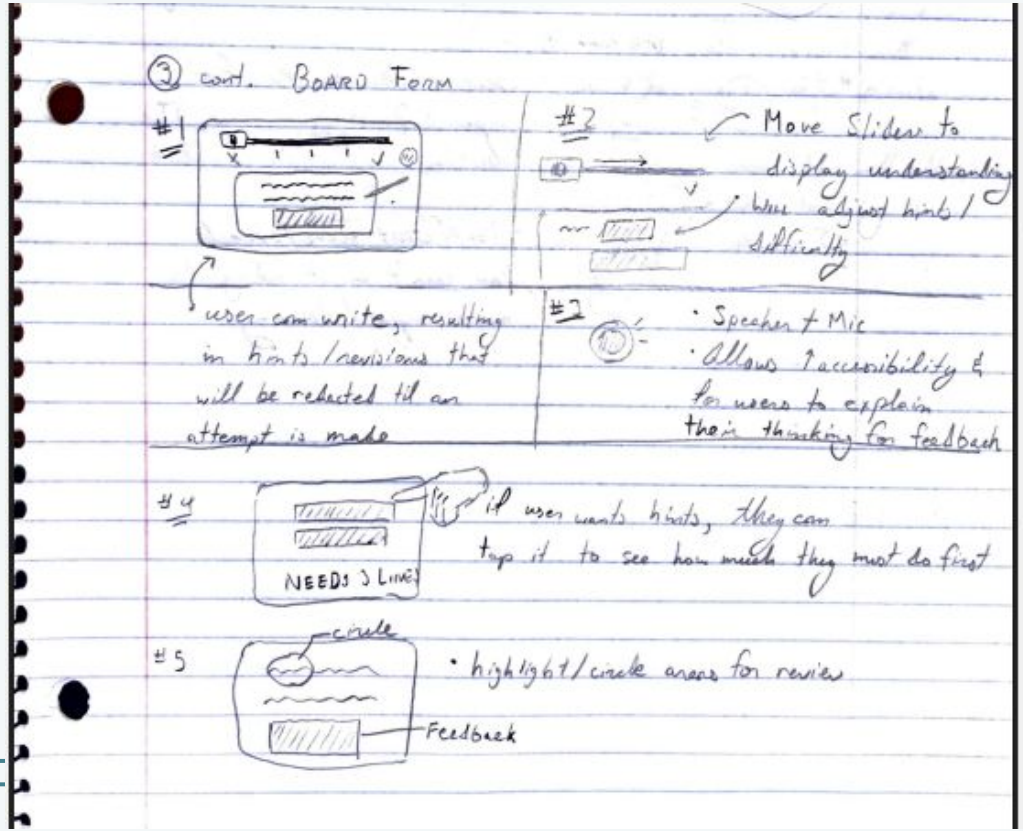
01

Sketches


Notepad
Blurred Sections
REDACTED SECTIONS
Speech to Text Assistant
Focuses on input text
SLIDER
SPEAKER
Beginner
Understand
DISPLAY
SCAN + LIGHT
OR DESK LAMP
Building Block
explanations on them
correct recording
piece it together
Vibration
of answer


Learning Pod
inside
sitting
include bonds
revisions + scenarios
Brain Monitor
AR goggles, commenting on your attempt

Top 3 Ideas & Realizations



(4)

#1

- Lamp will first allow user an attempt while scanning

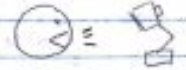
#2

- once movement stops + hand moved start scanning

#3

- if sufficient work, annotate + add comments
  - ▭ = highlight errors / changes

#4



- speak to lamp for fast input / Questions via a TTS button

#5

- insufficient sense/speech can result in it asking for more input / thinking

(5)

#1

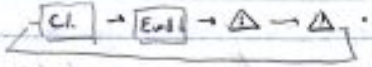
- Give blanks = pt of a thought
  - Ex. Q. 1, 7+7=5 → Not = ?
  - blanks via a ? side

=, ≠, ?, ~, CUSTOM

#2

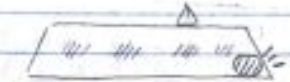
- chain a thought together

#3



- Any blanks on △ will be areas that need help

#4

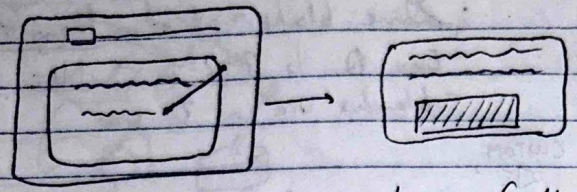


- if sufficient blanks placed + thinking done, board will begin asking in filling blanks

#5

- if a blank uses ▭ will allow a custom input / thinking to deal w/ more topics

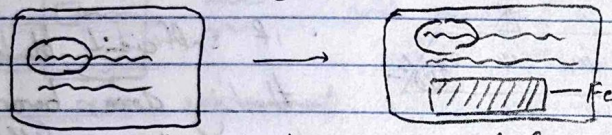
02 Frameworks



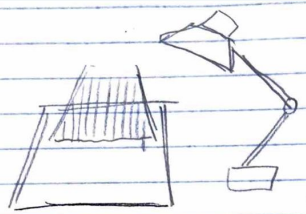
User writes in display

begin feedback at a stopping point/ submission

- User move slider right for ↑ understanding
- w/ ↑ understanding the board will reduce hints / interventions

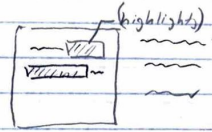


- User can circle content to get feedback
- reveal feedback based on circle

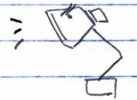


Lamp scans content after a break

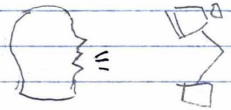
IF sufficient, begin annotating



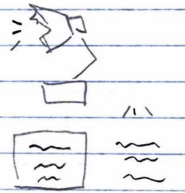
IF INSUFFICIENT WORK lamp will request more thinking or ask probing questions



If speaking to the lamp



Can act as another input, giving annotations on speaker output



03

Selected Interface

Why? Hopefully you’re not Board yet.

Pros:

- Forces Active learning
- Interactive
- Lightweight & Draggable
- Incremental hints

Cons:

- Could still be abused
- Understanding may be subjective
- User feedback may be abstract
- Display may be inconvenient for some websites

Why? Lamporghini.

Pros:

- Forces Active learning
- Student must write or construct ideas first
- AI reacts to reasoning, not just final answers
- Incremental hints
- Encourages revision cycles

Cons:

- Image detection may not work under some circumstances (i.e. blurry photoes)
- “Sufficient” may lack accurate measurement
- Uses more physical technology
- If reasoning is unclear or poorly written, AI may misinterpret what the student is saying

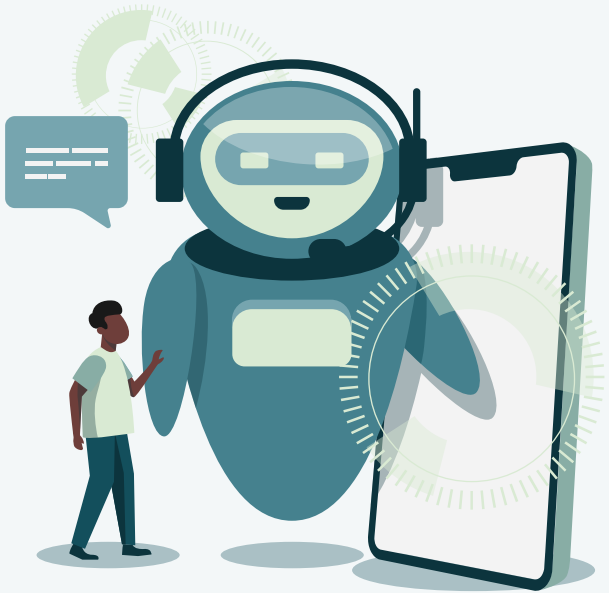
Why (Design Process)?

- • **86% of students** report using AI tools for schoolwork at least once
- • **~40–50% admit** they sometimes copy AI responses without fully understanding them
- Students retain significantly more info when actively constructing info rather than reading/listening to them
- In our G1-G2 interviews multiple stated that they wanted “independence” but “defaulted” to the convenience of AI

This implies that the problem with AI is the lack of passive learning rather than the tool itself

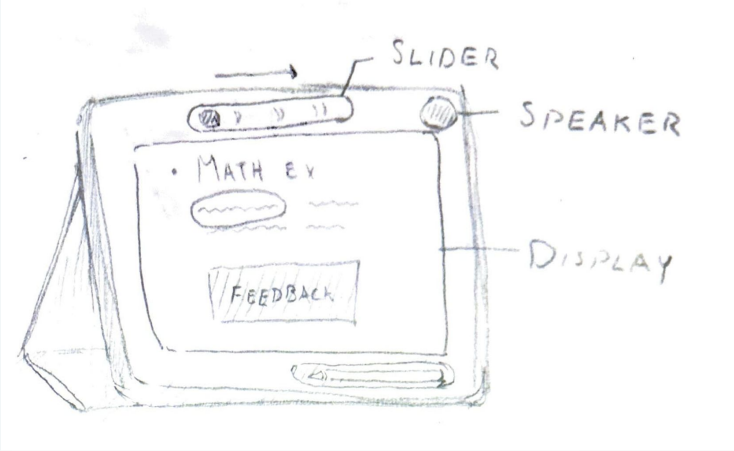
04

Low-fi prototype construction



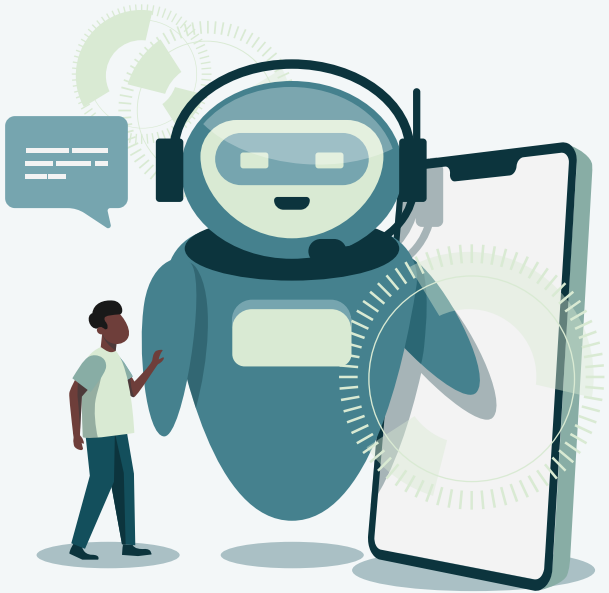
Features + Interactions

- Board
  - Reason Lab gives you a board where you write your reasoning
  - The AI annotates directly on your work
  - You revise on the same board Feedback updates dynamically It feels like a whiteboard session with a tutor.
- Slider
  - Before receiving hints, the user adjusts a difficulty slider that controls how much guidance they receive.
  - Instead of AI deciding everything, the student chooses the challenge level, reinforcing agency and metacognition.
- Speaker / Mic
  - Text-to-Speech (TTS): Instead of typing, the student must verbally explain their reasoning.
  - ReasonLab listens, analyzes understanding, and responds with adaptive spoken prompts.



05

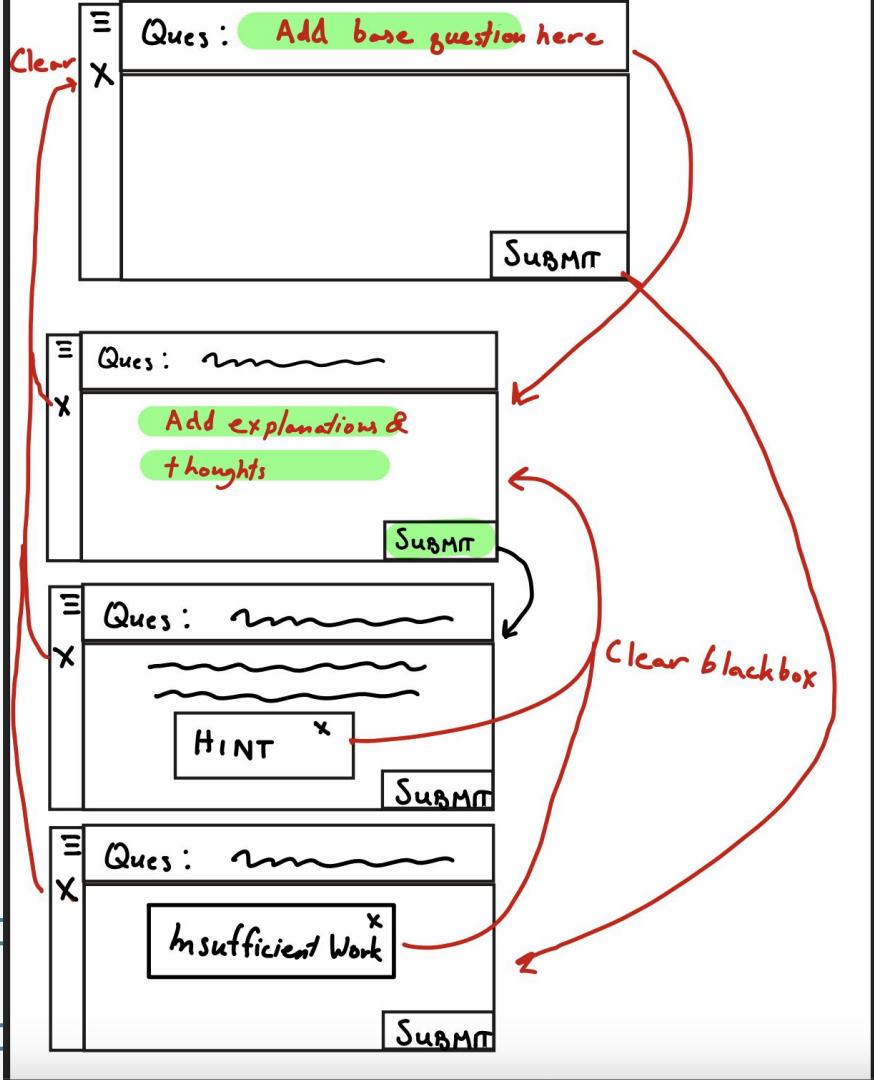
Low-Fi Prototype



Task Flows

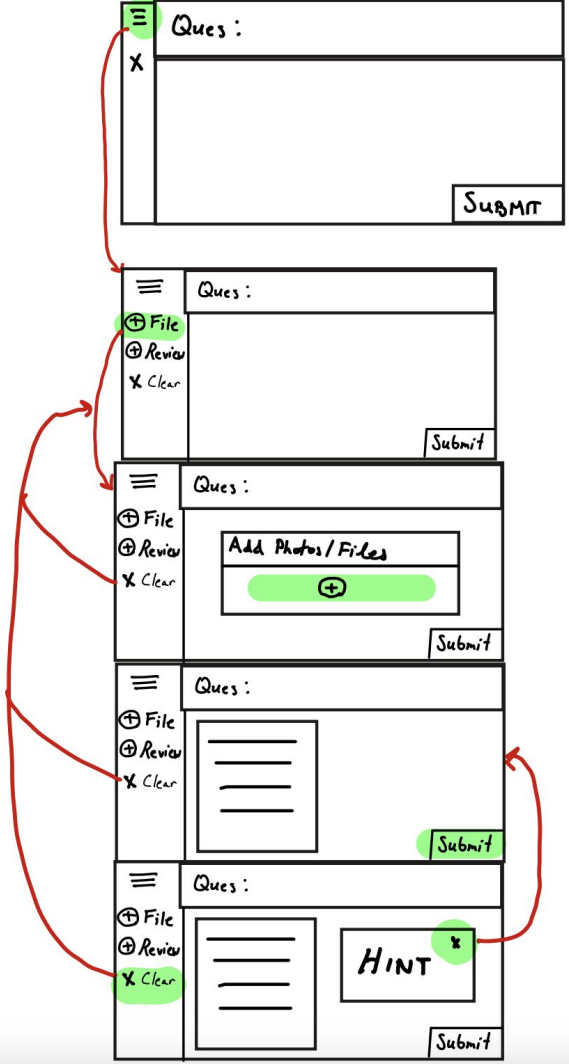
Task 1- Simple

Get a Hint



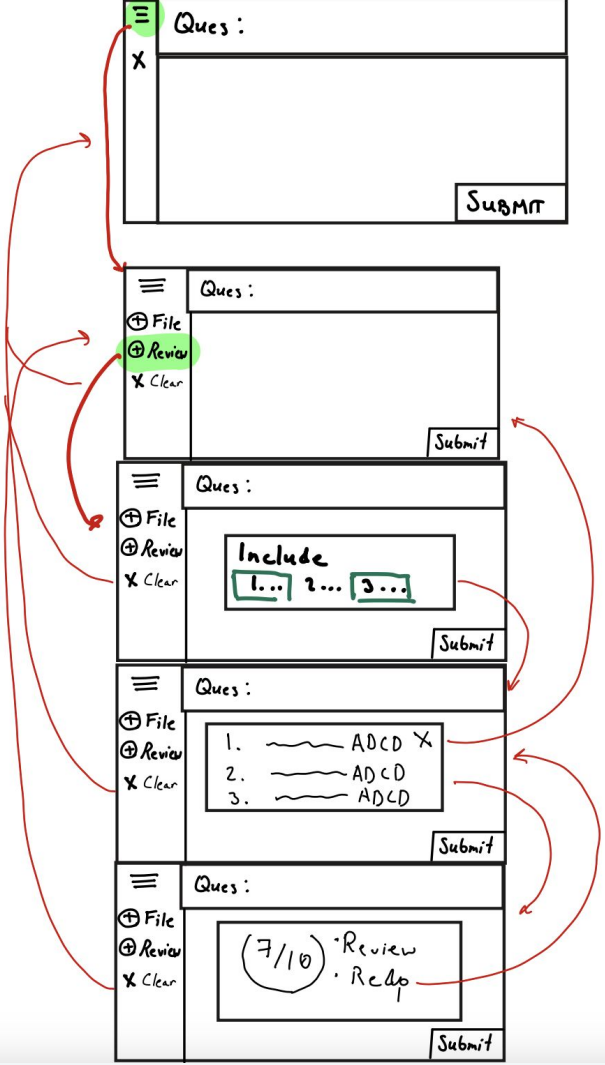
Task 2- Moderate

Import a File



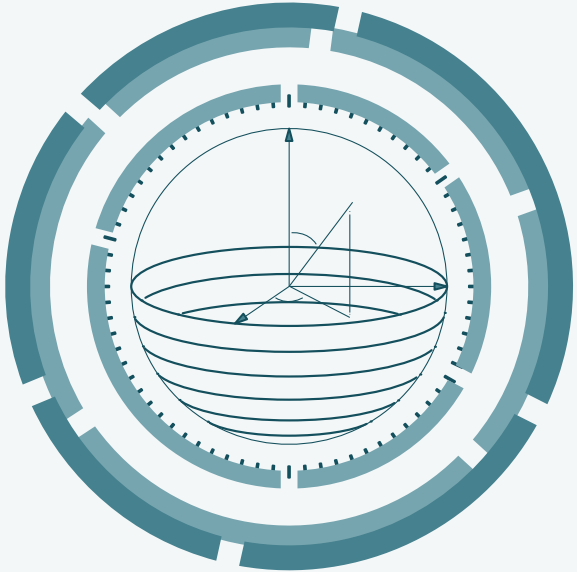
Task 3 - Complex

Get the AI to Create a Specific Study Guide



06

Testing Methodology



Participants

Steve is a 19-year-old college freshman, Humanities-focused, struggles with structuring arguments. Met in the Student Resource Building.

Alex is a 20-year-old college sophomore, recently switched majors, learning unfamiliar material

Recruited via campus email list.

Entity 303 is a 21-year-old college senior, STEM-focused, frequent AI user. Met them in another STEM class.

Environment + Apparatus

Test: Steve

Environment:

- Quiet study room in college library
- Individual desk with minimal distractions

Apparatus:

- Laptop running ReasonLab board & slider interface
- Screen recording software to capture interactions
- Notebook for self-notes

Test: Alex

Environment:

- High school classroom after school

Apparatus:

- Tablet with stylus for handwriting/revision
- ReasonLab interactive board prototype

Test: Entity 303

Environment:

- High school classroom after school

Apparatus:

- Tablet with stylus for handwriting/revision
- Paper for sketching rough drafts before submitting

Procedure

Facilitator: Junjie

- Introduce the AI Writing Board
- Demo interaction mechanics
- Explain each task (Simple → Moderate → Complex)
- Allow participants to complete each task independently
- Ask participants to rate mental effort
- Ask for general feedback
  - what was confusing?
  - what was helpful?
  - what can be improved?

Notetaker: Anthony

- Track the time taken per task
- Write down any key issues the participant is running into
- Note down feedback

Emulator: Ayala

- Replicate the behavior of the AI
- If feedback is requested, evaluate if enough work was done, and help accordingly

Goals + Key Measurements

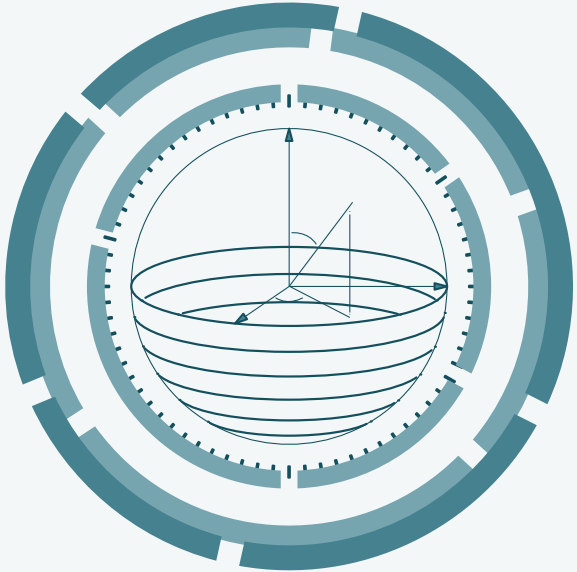
Goal 1: Intuitive

- We want the board/teaching assistant to aid the user rather than be against them
- Will need to **track hesitation / time spent getting AI feedback** rather than working to get a better understanding of whether the AI is actually helpful
- We will also **ask directly** again what they think is **helpful or confusing**

Goal 2: Low interface complexity

- **Time to find** certain features
  - ex. How long to find the Study Guide + File import features
- Non educational items shouldn't cause any mental strain. Should be intuitive and known to all the users.

07



Test Results

Processed Data

Notes across 3 participants

- Felt "stretched in a good way" — had to pause and reflect
  - All users hesitated at explain-back step (avg ~10 seconds pause)
- Sometimes unsure what the AI was “looking for” in the summary
  - 2/3 asked what counts as “good enough” explanation
- All used the complexity slider at least once
  - Appreciated depth control, helped “slow down and focus”
- Getting a quiz at the end was satisfying and it felt like closure
- No one abandoned the task

Big Picture:

- The explain-back step increases cognitive engagement.
- Users are willing to accept friction if it improves learning.
- Slider feature is intuitive enough to encourage exploration.
- Some confusion exists around explanation expectations.

Key Measurements

Goal 1 – Intuitive Learning Support

- Avg mental effort: **4.0 / 5**
- Avg helpfulness rating: **4.3 / 5**
- Completion rate: **100%**
- Abandonment Rate: 0%

Goal 2 – Interface Complexity

- Avg time to find slider: **6 seconds**
- Avg time to find Study Guide: **9 seconds**
- Avg misclicks: **1 per participant**
- Interface clarity rating: **3.8 / 5**

This means…

- Cognitive effort is high but perceived positively.
- Interface complexity is acceptable but needs refinement.
- System supports learning without overwhelming users.

Other Relevant Observations

Pros:

- Participants verbalized reasoning more during explain-back.
- Lean-in body posture during quiz stage.
- Users expressed satisfaction after confirming understanding.
- “This makes me slow down in a good way.”

Con:

- Uncertainty about what qualifies as “good enough” explanation.
- Mild frustration when hint limit was reached.
- One participant hesitated before adjusting slider.

Usability Goal Evaluation

Goal 1: Intuitive Learning Support

Status: Mostly Achieved

- Users felt AI was helpful, not adversarial.
- Mental effort was high but interpreted as productive.
- No abandonment occurred.
- Needs improvement: clearer explain-back expectations.

Goal 2: Low Interface Complexity

Status: Partially Achieved

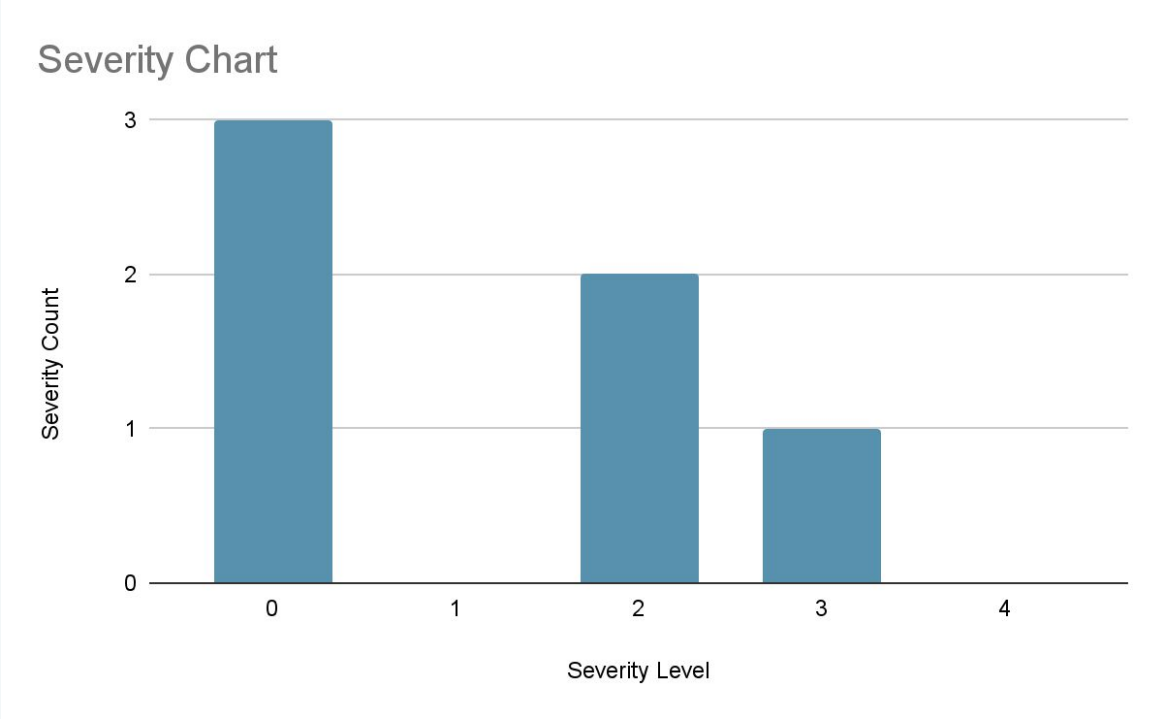
- Features were discoverable within acceptable time.
- Minor confusion about slider meaning.
- Navigation added slight but manageable friction.

Critical Incidents

Incident	Type	Severity (0–4)	Evidence	Justification
Confusion about explanation threshold	Negative	3	2/3 participants asked what counted as “good enough” explanation	Affects core learning flow; interrupts main interaction
Slider label ambiguity	Negative	2	Users hesitated before adjusting slider	Minor but repeated confusion about feature meaning
Frustration at hint limit	Negative	2	Participants verbalized mild frustration when hints ran out	Did not block task completion, but impacted experience
Increased engagement during quiz	Positive	0	Participants leaned forward and focused during quiz	Desired reflective behavior; no usability issue
Voluntary retry after quiz failure	Positive	0	Participant retried immediately without prompting	Strong signal of maintained agency
Used slider proactively	Positive	0	2/3 participants increased complexity after first success	Indicates adaptive feature was intuitive

No catastrophic failures, core concept validated, refinement required

Chart



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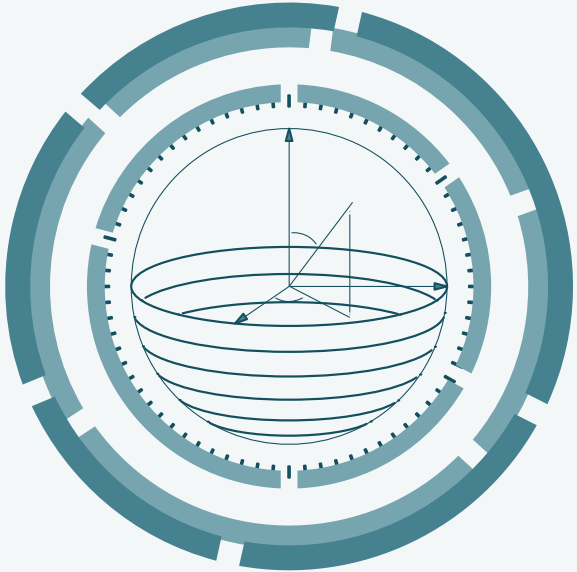


Discussion

Implications & Design Changes

Immediate Design Changes:

- Add example sentence starters for explain-back.
- Add visual indicator of hint usage limit.
- Clarify slider levels with descriptors (Beginner / Guided / Advanced).
- Improve post-quiz feedback clarity.

What this means:

- The reflective learning mechanism is effective.
- Friction is tolerable when learning value is clear.
- Interface clarity improvements will reduce cognitive overhead.
- System is conceptually strong but needs UX refinement

Variations

Differences observed across participants

- (P1): More hesitation at explain-back.
- (P2): More sensitive to UI labeling.
- (P3): Most comfortable with structured friction.

Validity Considerations

- Small sample size (n=3).
- All participants moderately self-motivated learners.
- Real academic stakes were not present.
- Long-term habit change not measured.

Applicability

Results likely generalize to:

- Self-driven learners
- Technical or structured learning environments

Further testing needed for:

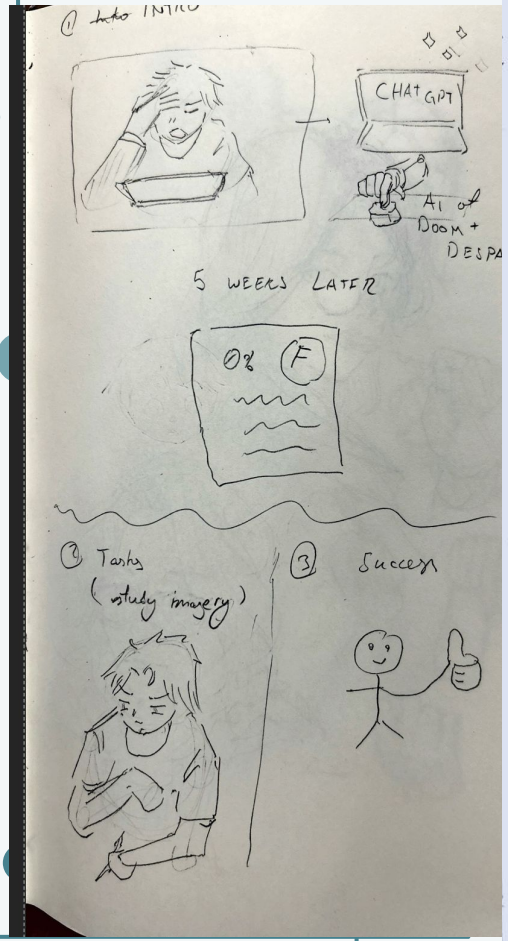
- Users who rely heavily on shortcut-based AI
- Users with lower intrinsic motivation

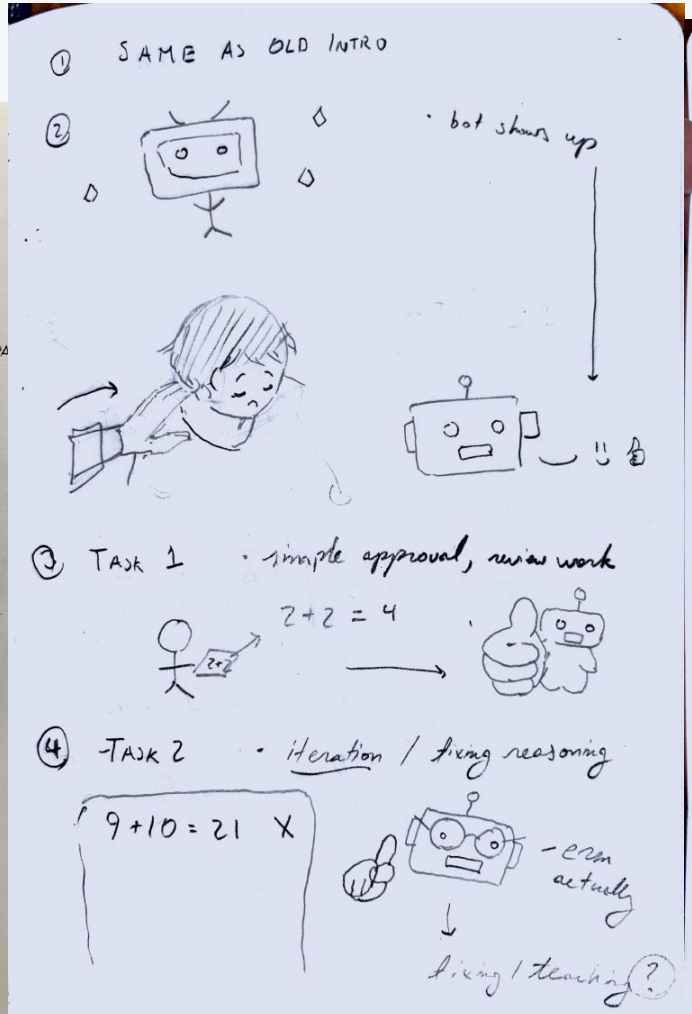
Overall Conclusion

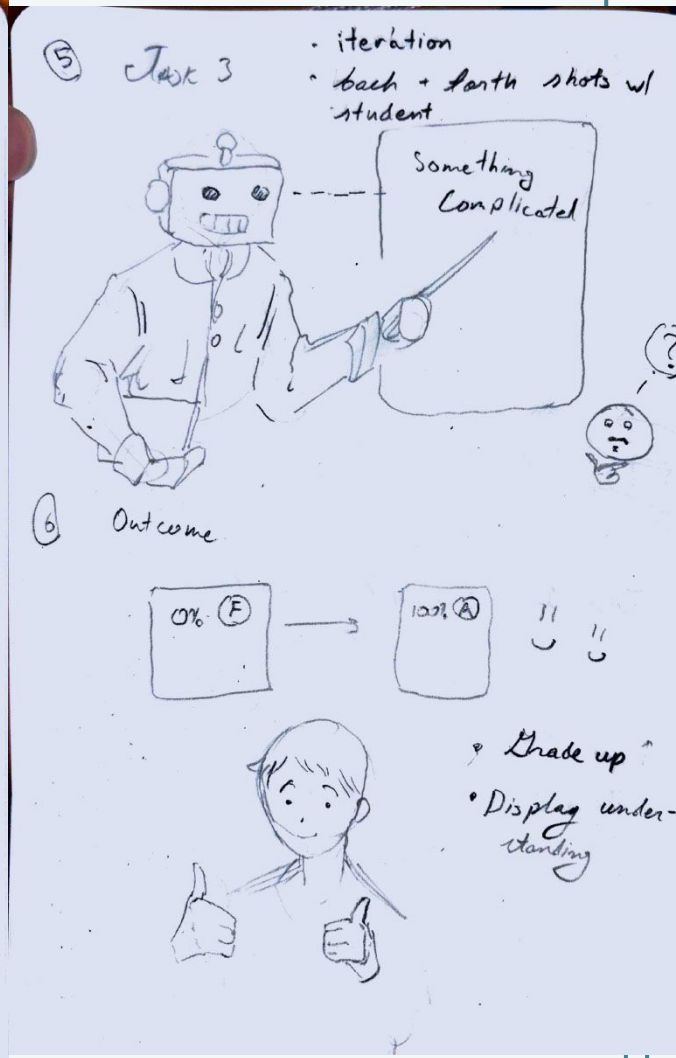
What We Learned

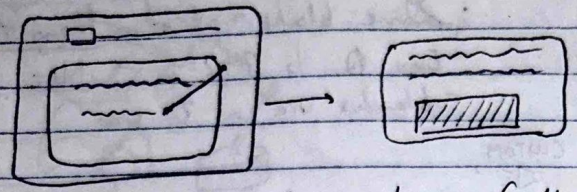
- The system successfully promotes reflective engagement.
- Explain-back is powerful but needs scaffolding.
- Interface is mostly intuitive with minor friction.
- No major structural failures occurred.

08 Appendix


① Intro INTRO
CHAT GPT
AI of Doom + DESPA
5 WEEKS LATER
0% F
② Tasks (study imagery)
③ Success


① SAME AS OLD INTRO
② · bot shows up
③ TASK 1 · simple approval, review work
2+2 = 4
④ TASK 2 · iteration / fixing reasoning
9+10 = 21 X
- erm actually
fixing / teaching ?


⑤ Task 3
· iteration
· back + forth shots w/ student
Something Complicated
⑥ Outcome
0% F
100% A
· Grade up
· Display understanding



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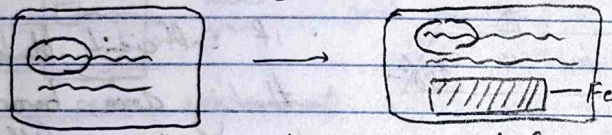

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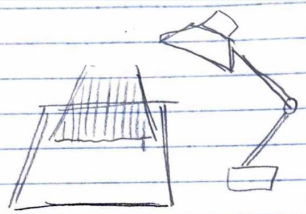

Lamp scans content after a break

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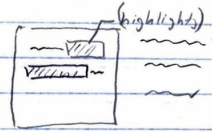

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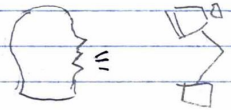

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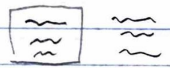
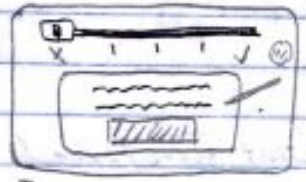


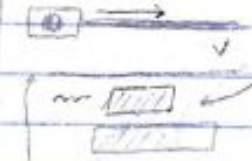
annotation via light

③ cont. BOARD FORM

#1



#2



Move Slider to display understanding

Will adjust hints / difficulty

user can write, resulting in hints / revisions that will be reflected til an attempt is made

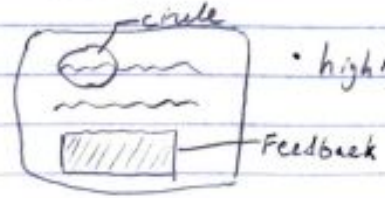
#3

- Speaker + Mic
- Allows accessibility & for users to explain their thinking for feedback

#4

if user wants hints, they can tap it to see how much they must do first

#5



- highlight/circle areas for review

(4)

#1

- Lamp will first allow user an attempt while scanning

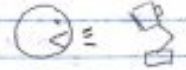
#2

- once movement stops + hand moved start scanning

#3

- if sufficient work, annotate + add comments
  - ▭ = highlight errors / changes

#4



- speak to lamp for fast input / Questions via a TTS button

#5

- insufficient sense/speech can result in it asking for more input / thinking

(5)

#1

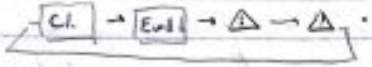
- Give blanks = pt of a thought
  - Ex. Q. 1, 7+7=5 → Not = ?
  - blanks via a ? side

=, ≠, ?, ~, CUSTOM

#2

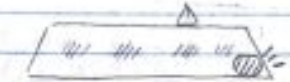
- chain a thought together

#3



- Any blanks on △ will be areas that need help

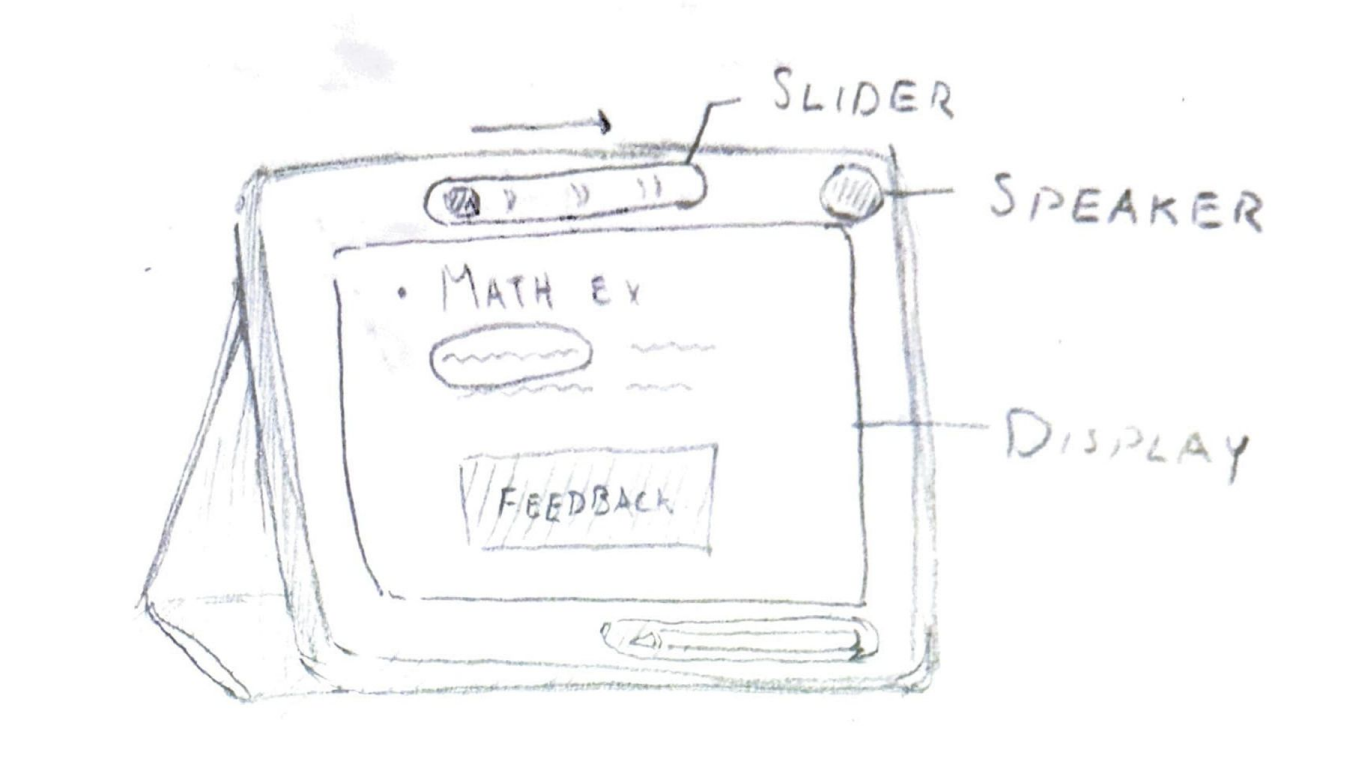
#4

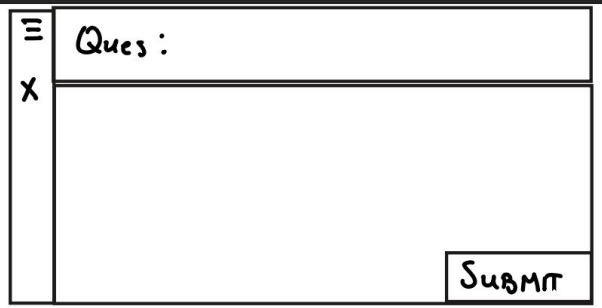


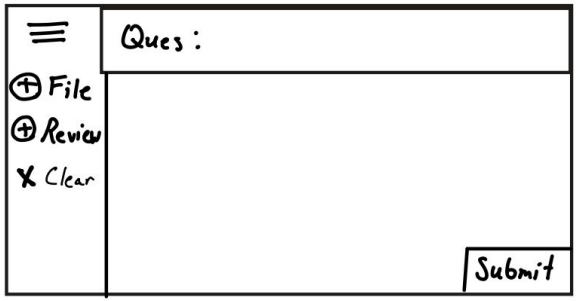
- if sufficient blanks placed + thinking done, board will begin asking in filling blanks

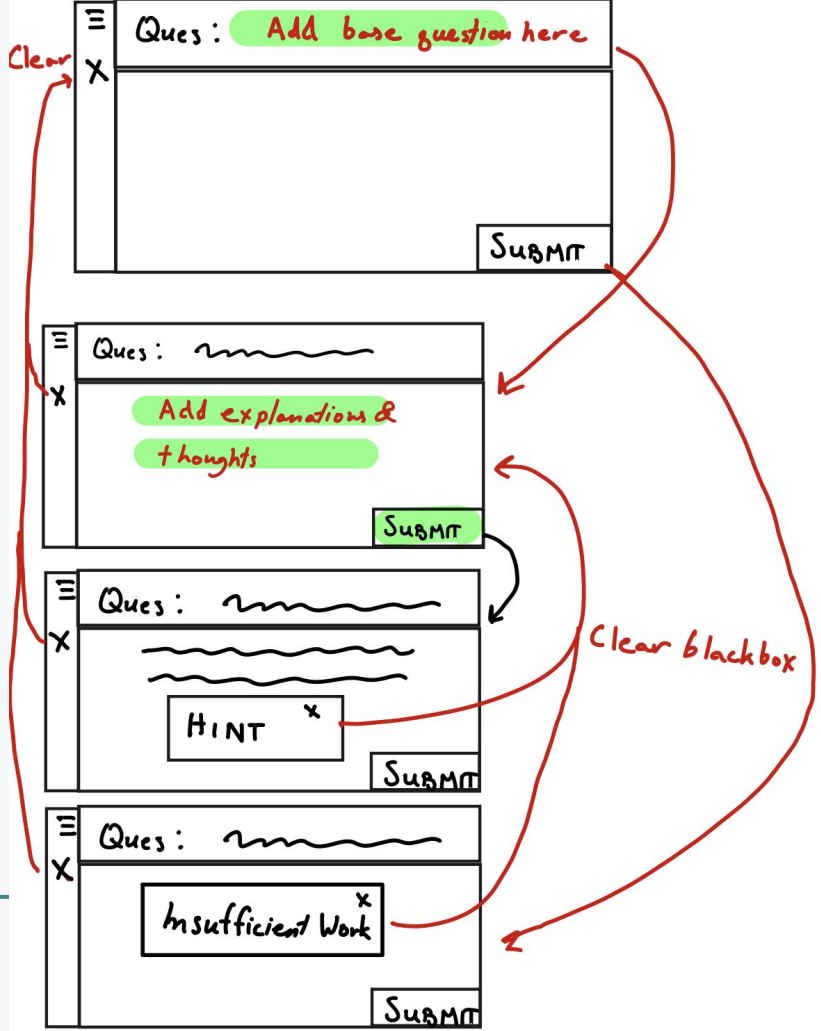
#5

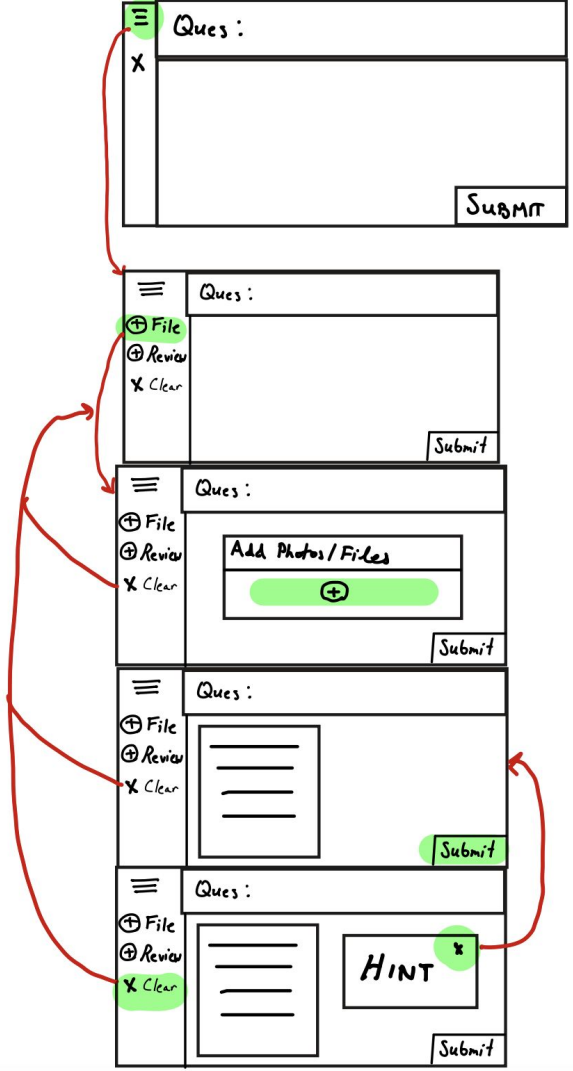
- if a blank uses ▭ will allow a custom input / thinking to deal w/ more topics

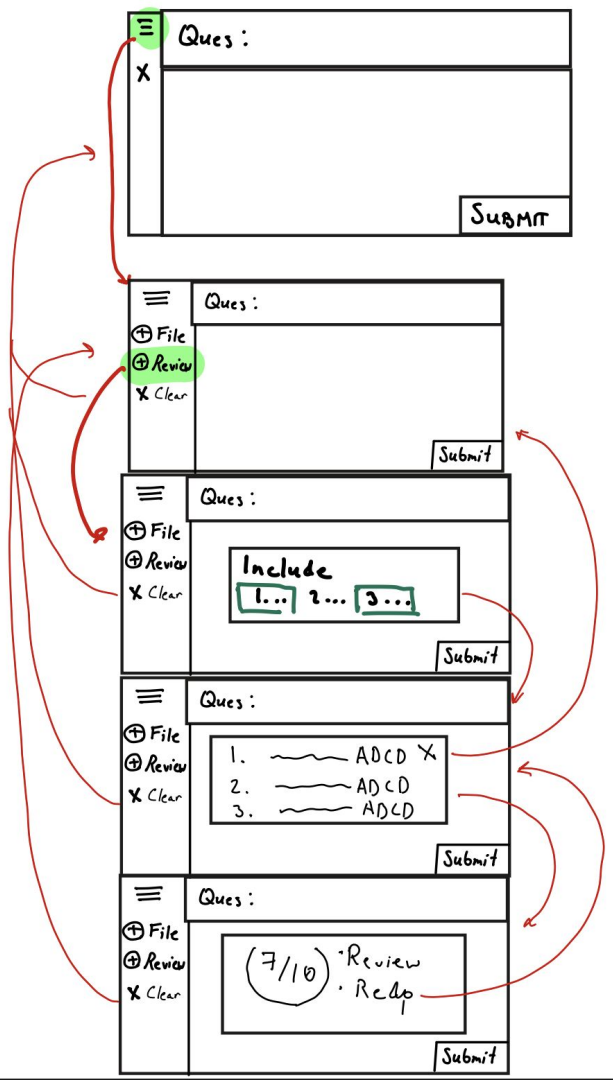

SLIDER
SPEAKER
MATH EX
DISPLAY
FEEDBACK


Ques :
SUBMIT


Ques :
File
Review
Clear
Submit


Clear
Ques : Add base question here
SUBMIT
Ques :
Add explanations & thoughts
SUBMIT
Ques :
HINT
SUBMIT
Clear blackbox
Ques :
Insufficient Work
SUBMIT


Ques :
X
SUBMIT
Ques :
File
Review
Clear
Submit
Ques :
File
Review
Clear
Add Photos / Files
Submit
Ques :
File
Review
Clear
Submit
Ques :
File
Review
Clear
HINT
Submit


Ques :
Submit
File
Review
Clear
Ques :
Submit
File
Review
Clear
Ques :
Include
1...
2...
3...
Submit
File
Review
Clear
Ques :
1. ADCD
2. ADCD
3. ADCD
Submit
File
Review
Clear
Ques :
7/10
Review
Recap
Submit